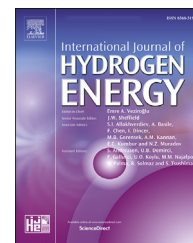


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Economic accounting and high-tech strategy for sustainable production: A case study of methanol production from CO₂ hydrogenation

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HIGHLIGHTS

- Introducing a new methanol synthesise plant from CO₂ hydrogenation.
- Techno-economic assessments of the introduced plant and propose the construction idea of high-tech park.
- The sustainable-based methanol synthesise is compared to the conventional methanol production method.

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ABSTRACT

The global proposal of ‘carbon neutrality’ puts forward higher innovation demand for the cleaner energy production. The potential for employing “green” methanol produced from hydrogen obtained by water electrolysis and collected CO₂ from a gas-fired power station is examined in this study.

The consumption of electricity for renewable methanol production is 1.045 times as much as that for traditional methanol production, the traditional method consumes 2.5 times as much thermal energy as the renewable methanol process. In addition, the total direct and indirect CO₂ emissions from renewable methanol production are almost one-third of the emissions from the traditional method. The total cost of setting up the units of a renewable and a traditional methanol production plant with an annual capacity of 100,000 tons is \$50.1 million and \$46.806 million in this study case, respectively. If the methanol price hits \$310 per ton, renewable methanol production will be highly economically viable. But if electricity and gas prices rise or CO₂ emission tax is imposed, renewable and conventional methanol production plants will lose their economic feasibility. Therefore, in order to deal with this risk, the establishment of special high-tech parks is of great significance to reduce costs and stabilize the sustainable development of relevant industries.

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Introduction

Exploring to find more green solutions for solving environmental issues has captured attention of experts and researchers [1–4,58]. On the other hand, producing energy and drinking water for residents with respect to the environmental parameters has been become a challenges for policy makers, too [5–9]. In recent decades, fossil fuel combustion for electricity production and vehicle fuels has considerably increased the level of CO₂ emissions, leading to a surge in the global average atmospheric CO₂ concentration from 280 parts per million (ppm) before the industrial revolution to 370 ppm at present [10,11]. A significant amount of fossil resources is used for conventional methanol production based on synthesis gas (syngas). This produces considerable carbon dioxide emissions, i.e. each ton of methanol produced with this method releases 0.6–1.5 tons of CO₂ into the air [12,13]. Renewable methanol is produced from carbon dioxide emitted from steam power plants and hydrogen produced by renewable electricity. Renewable hydrogen is produced in three ways: water electrolysis, biomass conversion, and solar energy conversion [14]. In the water electrolysis method, the H₂ and O₂ are separated using renewable energy sources (RES) such as photovoltaic (PV), wind, hydroelectric, and geothermal power plants, as well as fossil sources. Another method for hydrogen production is using thermochemical processes such as gasification and pyrolysis or biochemical conversion of biomass. In the solar energy conversion method, through reactions such as thermolysis and photolysis, thermal energy and photons are respectively used to synthesize hydrogen [15,16].

This should be included when determining the actual greenhouse gas reduction potential of CCU. In this regard, a life cycle assessment (LCA) is critical for determining the capacity of each CCU plant to avoid greenhouse gas emission [8]. In Ref. [9] authors emphasize that the quantity of CO₂ that is utilized is not the same as the amount of carbene dixide that is avoided. The overall balance should account for all value chain echelons: CO₂ capture, transport, transformation, and consumption of carbene dioxide products. However, the primary benefit of a plant that utilizes captured carbene dioxide, referred to as a The CCU-plant will continue to reduce feed-stock usage from its inception. (typically a fossil fuel) and the avoidance of associated emissions, when compared to well-known and conventional pathway(s) for synthesizing a specific product. This may provide financial incentives for the implementation of CO₂ capture technology. Due to the intrinsic potential of CCU, particularly in terms of its usage to synthesize minerals, chemicals, and fuels, it must be used in conjunction with geological CO₂ storage if significant CO₂ emissions reductions are to be achieved [10]. Currently, existing CO₂ re-use accounts for 0.4% of carbon emitted [10].

A significant number of CO₂ capture patents (removal and separation) have been published recently (2006–2010), according to Li et al. [11]. However, capturing CO₂ from large, stationary sources remains a challenge due to concerns over safety, efficiency, and cost. There are distinctions between industrial and power plant CO₂ collection, as described by Roddy [12]. While power plants can use post-combustion

configurations to capture CO₂, industrial facilities must deal with retrofit layouts that are less direct, such as pre-combustion or oxy-fuel combustion. These studies show that integrated CO₂ capture-transport networks can be beneficial and interesting for both big and minor sources of CO₂. For CO₂ pipes, the network size will be determined by future advances in context, codes, and standards. A plethora studies [13–15] show how CO₂ may be used as a raw material at various stages of development and at various product scales and market potentials.

Increasing the carbon recovery conversion by catalytic synthesis is the most recent and cutting-edge technology. While electrochemical and photochemical conversion are relatively new, they have the potential to be more efficient and produce less CO₂. Formic acid synthesis and mineralisation can both be affected by pollutants in the CO₂ stream, depending on the technology used to create the final product from CO₂. This can have a substantial economic impact on both processes. As a strategic method for biodiesel production, alge synthesize has the significant role. Algae production is an emerging technology for biofuel synthesis that has the potential to make a significant contribution [16,17]. CCU is drawing policymakers' attention as a possible way to (i) stimulate local economies, (ii) manage CO₂ after capture, and (iii) potentially reduce CO₂ emissions. This is why CCU applications may include a variety of motivating factors.

Numerous recent investigations have been conducted on the synthesis of methanol using CO₂ hydrogenation. Joo et al. (1999) [12] investigated the two-step manufacture of methanol and concluded that it yields more than the one-step approach. Mignard et al. (2003) [18] presented a procedure for synthesizing methanol using CO₂ collected from coal power station flue gas and electrolytic hydrogen. To achieve significant CO₂ abatement, the process is dependent on the availability of waste heat from the power plant to give thermal energy to the process. Without these heat sources, CO₂ abatement is essentially non-existent. Mignard and Pritchard (2006) [19] examined the energy efficiency of CO₂ hydrogenation procedures for the manufacture of methanol, ethanol, and gasoline. The methanol procedure was found to be the most efficient. Pontzen et al. (2011) [20] conducted studies to compare the generation of methanol from CO₂ using a Cu/ZnO/Al₂O₃ catalyst to the typical syngas production procedure. When compared to the conventional technique, the CO₂-based one had lower productivities. Soltanieh et al. (2012) [21] investigated and analyzed the economics of co-producing methanol and energy from captured carbon dioxide and carbon-free hydrogen. Van Der Ham et al. (2012) [22] used a fluidised-bed membrane reactor to build a CO₂ to methanol process. Although the procedure resulted in significant CO₂ reductions, it was not economically viable.

The main novelty of this work:

- Currently, methanol production in energy-rich regions such as the Middle East is mainly through natural gas reforming, and to the extent of the author's knowledge and according to reliable sources, there is no renewable methanol plant in some countries such as Iraq even on a pilot basis. This is because they owns abundant proved natural gas reserves (or imported such as Russia-China

pipeline gas), making natural gas the main source of feedstock for methanol plants.

- Reduces the level of CO₂ in the air

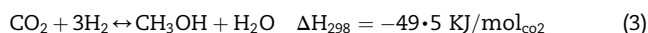
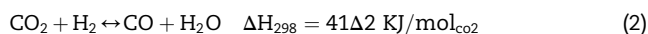
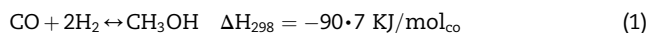
Provides a substance that can be used as an alternative fuel or as a starting material during organic synthesis and give reference for both energy-rich countries and main energy consumers such as China. The purpose of this study was to compare and contrast the conventional and renewable methanol production methods through a thermoeconomic analysis. In the former method, methanol is produced through natural gas reforming but in the latter method synthesis gas is produced from a compound of hydrogen from water electrolysis and CO₂ from a gas power plant and then it is converted into renewable methanol by going through a few stages. The methanol production from renewable sources was simulated using Lurgi technology and ASPEN HYSYS V.10 and then was economically analyzed using COMFAR III.

Methodology

ASPEN HYSYS V.10 was used to design and simulate the conversion of hydrogen and CO₂ into methanol with a 100,000 ton annual capacity (the standard commercial unit for methanol facilities). High-pressure streams were studied using the RedlicheKwong Soave equation of state (RKSMHV2), which incorporates modified HuroneVidal mixing rules. The NRTL-RK model was used for streams with low pressure (10 bar). This study involves the following steps:

General description of direct synthesis of methanol from CO₂

Methanol is produced by three-phase hydrogenation of CO₂ [23]:



The exothermic processes (Eq. (1)) and (Eq. (3)) yield methanol, while the endothermic reverse water-gas-shift (RWGS) reaction (Eq. (2)) happens simultaneously. Methanol and water are the products of methanol and hydrogen reaction. This is an equilibrium reaction. It is a catalytic reaction that takes place in a fixed-bed catalytic reactor. The catalyst used is usually Cu/ZnO/Al₂O₃ [17]. According to Vanden Busche and Froment's model [24], the reactions of methanol generation and RWGS may be accurately described using this catalyst. According to Sahibzada et al. (1998) [18], CO₂ is the principal carbon source for methanol production and does not directly block the reaction represented by Eq. (3), which is what the model assumes. The model also takes into account the inhibitory effect of the water generated during the RWGS process. By adjusting reaction activation energies, Mignard and Pritchard (2008) [25] improved the model's fit to experimental data while simultaneously extending the model's useable pressure range to 75 bar.

É.S.'s kinetic model is applied in this study. The Van-Dal and C. It's called Bouallou [26] Eqs. (4)–(6) represent the rearranged kinetic model, where pressure is expressed in Pa and temperature is expressed in K [26].

Methanol generation:

$$r_{\text{CH}_3\text{OH}} = \frac{K_1 P_{\text{CO}_2} P_{\text{H}_2} - K_6 P_{\text{H}_2\text{O}} P_{\text{CH}_3\text{OH}} P_{\text{H}_2}^2}{(1 + K_2 P_{\text{H}_2\text{O}} P_{\text{H}_2}^{-1} + K_3 P_{\text{H}_2}^{0.5} + K_4 P_{\text{H}_2\text{O}})^3} \left[\frac{\text{kmol}}{\text{kg}_{\text{cat}} \text{s}} \right] \quad (4)$$

RWGS reaction:

$$r_{\text{RWGS}} = \frac{K_5 P_{\text{CO}_2} - K_7 P_{\text{H}_2\text{O}} P_{\text{CO}} P_{\text{H}_2}^{-1}}{1 + K_2 P_{\text{H}_2\text{O}} P_{\text{H}_2}^{-1} + K_3 P_{\text{H}_2}^{0.5} + K_4 P_{\text{H}_2\text{O}}} \left[\frac{\text{kmol}}{\text{kg}_{\text{cat}} \text{s}} \right] \quad (5)$$

$$\ln K_i = A_i + \frac{B_i}{T} \quad (6)$$

Table 1 shows the Arrhenius law's parameters for the kinetic constants (Eqs. (4)–(6)).

Investigated the possibility of employing “green” methanol produced from RES as a value-added feedstock for a fully decarbonized chemical process industry as a sustainable alternative to the well-established technique of steam methane reforming. Water electrolysis is fed by energy recognized as renewable, such as electricity produced by a renewable energy cluster, grid curtailment (i.e. RES over-production), or low-carbon electric energy purchased from the grid in areas where RES penetration is very high. The system is assumed to have a constant supply of renewable electricity for water electrolysis; connecting the system with a real RES availability profile would necessitate a system of buffers for the raw gases (H₂ and, maybe, CO₂) that could be sized using a time-resolved model. The study focuses on commercial-scale methanol production (12,625.37 kg/h), and the size of the water electrolysis and carbon capture sections was chosen to feed the methanol synthesis section with adequate hydrogen and CO₂ collected fluxes. Set the capacity of the water electrolysis section to 10 MW, resulting in a continuous flow rate of 2778.09 kg/h of hydrogen at a pressure of 10 bar (specific consumption of 5.4 kW h/Nm³ H₂). The compressed electrolytic hydrogen (99.7% v without auxiliary purifying equipment) is then used to make methanol; oxygen is either vented or sold in the base scenario. A typical two-column carbon capture system recovers 17,704.33 kg/h of CO₂ (96.99% w

Table 1 – Parameters of the rearranged kinetic model [26].

K ₁	A ₁	–29.87
	B ₁	4811.2
K ₂	A ₂	8.147
	B ₂	0
K ₃	A ₃	–6.452
	B ₃	2068.4
K ₄	A ₄	–34.95
	B ₄	14928.9
K ₅	A ₅	4.804
	B ₅	–11797.5
K ₆	A ₆	17.55
	B ₆	–2249.8
K ₇	A ₇	0.131
	B ₇	–7023.5

Table 2 – The molar fractions of the input and output currents.

Name	N ₂	CO ₂	H ₂ O	MEA
Flue gas in	0.78	0.13	0.09	0
Sweet gas out	0.9493	0.0004	0.0502	0.0001
Absorbent out	0	0.0419	0.8608	0.0973
1	0	0.0229	0.8771	0.1
2	0	0.0419	0.8608	0.0973
3	0	0.0419	0.8608	0.0973
4	0	0.0252	0.8755	0.0993
5	0	0.0252	0.8755	0.0993
6	0	0.0253	0.8755	0.0992
7	0.0004	0.8551	0.1446	0
8	0.0004	0.8551	0.1446	0
9	0	0.0252	0.8755	0.0993
10	0	0.0252	0.8755	0.0993
Liq	0	0.0007	0.9993	0
Rich CO ₂	0.0004	0.9699	0.0297	0
Recycle	0	0.0253	0.8755	0.0992
Purg	0	0.0253	0.8755	0.0992
Make up	0	0	0.8925	0.0175

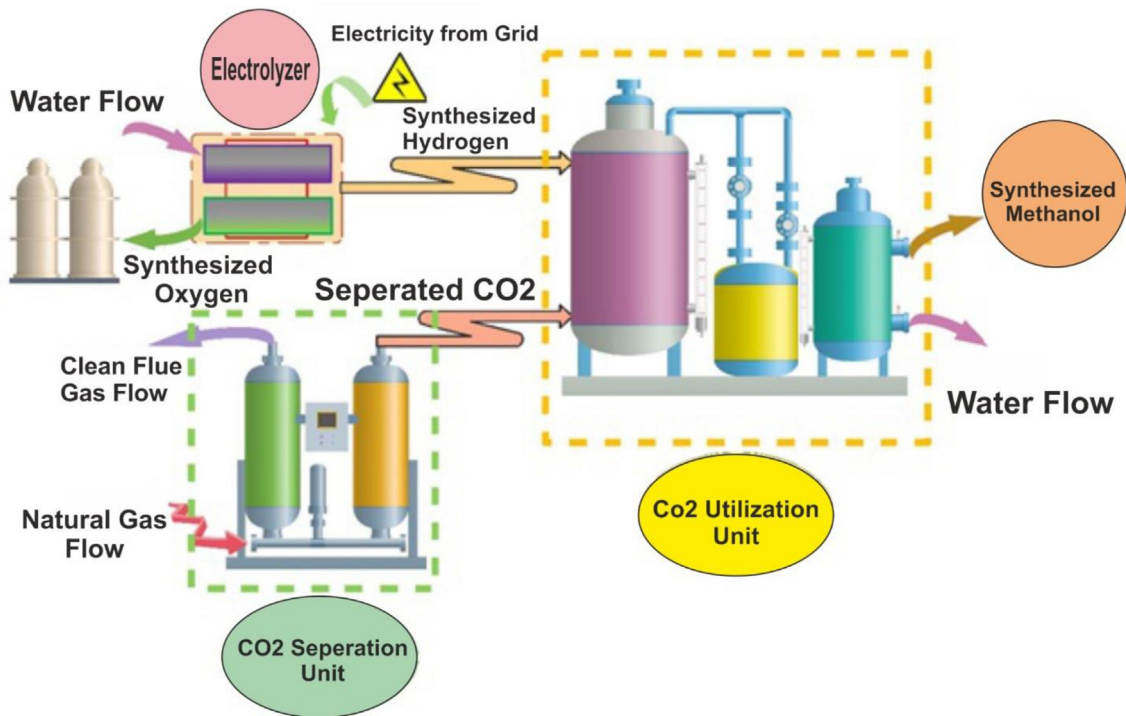
purity) from 93,270.96 kg/h of gas-fired power plant exhausts using an amine scrubbing procedure with an aqueous phase of Monoethanolamine (MEA) is the most well-established technology on the market.

Carbon dioxide is converted into methanol through one and two-step processes. The one-step CO₂-methanol conversion process is carried out by direct CO₂ hydrogenation but in the two-step process, the CO₂ is first converted into CO

through the reverse water gas shift (RWGS) reaction, and then it is converted into methanol by hydrogenation [12]. In this study, the one-step CO₂-methanol conversion process is utilized. The process flowsheet is presented in Fig. 1.

CO₂ is captured and mixed with electrolysis hydrogen from the water electrolysis stage in an intercooled compression train. The heated H₂/CO₂ feed (65 bar) is fed to the reactor for methanol synthesis via direct CO₂ hydrogenation: the state of the art consists of a reactor equipped with a copper-based catalyst – CuO/ZnO/Al₂O₃ – working in the pressure range of 50–100 bar [27,28]. CuO/ZnO/Al₂O₃ is the most frequent catalyst for both CO₂ hydrogenation and classical methanol synthesis from CO-rich syngas, as it is particularly active in both processes. The operating pressure was chosen because the Cu-based catalyst requires a pressure level of 5–10 MPa to achieve good methanol selectivity (above 50%) in the presence of Carbon dioxide and H₂, whereas at lower pressure, the rate of formation of CO by reverse water gas shift becomes higher than that of methanol, resulting in extremely low CH₃OH selectivity [29]. The gas combination exiting the reactor goes through some pressure drop and temperature swing phases to flash separate gases other than methanol, then a distillation column separates the methanol, and finally a mechanical separator raises the purity of the generated stream to 99.9% CH₃OH.

ASPEN HYSYS V.10 was used to investigate the process and size the devices. The pumps and flash drums have adiabatic functions. The efficiency of the compressors is considered to be 70%. The amount of recycled CO₂ is 17704.33 kg/h, which is the basis of the simulation and economic calculations.

**Fig. 1 – The schematic of methanol synthesis from CO₂ Hydrogenation**

Environmental impact assessment

Environmental factors include direct CO₂ emissions resulting from the process per se and indirect CO₂ emissions caused by electricity consumption [36–39]. The overall efficiency of carbon conversion is calculated according to Eq. (7) [40,41].

$$C_c = \frac{m_{\text{MeOH}_{\text{out}}}}{m_{\text{CO}_{2\text{in}}}} \cdot 100 \quad (7)$$

where $m_{\text{CO}_{2\text{in}}}$ and $m_{\text{MeOH}_{\text{out}}}$ represent input CO₂ molar flow rate and produced methanol molar flow rate, respectively.

Energy efficiency

Energy efficiency is calculated with Eq. (8) [31].

$$\mu = \frac{m_{\text{MeOH}} \text{LHV}_{\text{MeOH}} + \text{elec}_{\text{out}}}{\text{elec}_{\text{in}}} \cdot 100 \quad (8)$$

where μ represents energy efficiency, m_{MeOH} represents methanol mass flow rate, LHV_{MeOH} is the lower heating value of methanol, elec_{in} is input electricity, and elec_{out} is output electricity.

Economic assumptions

The economic analysis performance indicators evaluated in this study include the investment Cost of building the methanol plant, operational methanol production costs, payback period, and the Internal Rate of Return (IRR). Economic analysis of processes is based on the total annual costs which are usually used in conceptual designing of chemical processes. Furthermore, in this study, the investment cost and economic profitability of the conventional methanol production from natural gas (Unit 1) and renewable methanol production (Unit 2) are compared under each of the optimistic, realistic, and pessimistic price scenarios. COMFAR III was used to calculate the Internal Rate of Return (IRR), total investment cost, and total production cost. This software is among the most reliable analytical software for economic and financial analyses of investment projects.

Assumptions

The economic assumptions on which this study is based are as follows [32–35]:

- The lifespan of the plant is 20 years.
- The construction period of a petrochemical plant is assumed to be 2 years and the plant is assumed to become operational in 2023.
- The Depreciation is calculated using the straight-line depreciation method.
- The Salvage Value of the equipment is set at 10%.
- The average working hours for the methanol plant is estimated at 7920 h per year.
- The minimum discount rate is considered 18%.
- The average Income Tax is assumed to be 9%.

Investment cost

The Investment Cost is estimated by considering a number of factors and indicators. Eq. (9) is used for this purpose [36].

$$C_2 = (n^e)(C_1) \left(\frac{S_2}{nS_1} \right)^f \quad (9)$$

where the cost of a process unit is C_2 , the number of equally-sized units is n , the cost of a reference unit is C_1 , the capacity of a process unit is S_2 , the capacity of a reference unit is S_1 , the cost scaling exponent is e , and the cost scaling factor is f are all costs. All machinery and equipment except compressors are costed using this calculation. For compressors, the compressor capital cost estimation equation is used. The estimated costs of processing units are taken from various sources and various years, so Eq. (10), which is formulated according to the Chemical Engineering Plant Cost Index (CEPCI), should be used to estimate the current investment cost for the processing units [44–46].

$$C = C_2 \frac{i}{i_2} \quad (10)$$

In this equation, i represents the current value of CEPCI and i_2 is the value of CEPCI at the time of the determination of the reference cost [47,48]. Each of the two methanol production processes is divided into several parts and the total Investment Cost of the project is shown as the sum of the costs of individual parts.

Production cost

The cost of methanol production (ton/USD) is calculated with Eq. (11) [35,38].

$$C_{\text{MeOH}} = \frac{C_{\text{opt}} + C_{\text{dep}} - C_{\text{byp}}}{P_{\text{MeOH}}} \quad (11)$$

where P_{MeOH} is the methanol production capacity (tons/year), C_{byp} is the revenue generated from the sale of byproducts (electricity, oxygen, CO₂), C_{opt} is the operational cost, and C_{dep} is the depreciation cost.

Payback period

The payback period is calculated with Eq. (12) [38].

$$t = \frac{C_{\text{inv}}}{C_{\text{REV MeOH}} + C_{\text{byp}} - C_{\text{opt}}} \quad (12)$$

where t represents the time of return, C_{inv} represents the cost of investment, $C_{\text{REV MeOH}}$ is the annual revenue from methanol sales, C_{byp} is the byproducts sales revenue, and C_{opt} is the operational cost, including labor force, biomass, catalysts, and electricity.

Internal Rate of Return (IRR)

The IRR is used for a more realistic assessment of the productivity of a project [30,50], which actually offsets the discount rate (i) of the project's future net cash flows against its initial capital cost [42,43,49–51], as shown in Eq. (13). In other words, the Internal Rate of Return is the discount rate that offsets the Net Present Value (NPV) [39].

$$\sum_{t=0}^N (CI - CO)_t (1 + IRR)^{-t} = 0 \quad (13)$$

In this equation, IRR is the Internal Rate of Return, CI is the input cash flow in year t , CO is the output cash flow in year t , and N is the project's lifespan (20 years). In order to assess the feasibility of the project, the discount rate should be set at i_0 . If IRR is bigger than i_0 ($IRR > i_0$), it means that the project is economically feasible, or else it has no economic viability.

Simulation

CO₂ absorption process simulation with MEA solvent

The process is designed based on MEA to collect 17,704.33 kg/h of CO₂ (140,218 kg/year) from a gas-fired power plant (Fig. 2).

The physical properties of the CO₂ amine-based solution are estimated with the Kent-Eisenburg method. This method provides VLE and enthalpy values. The combustion gas (containing 78% v/v nitrogen, 13% v/v CO₂, 9% v/v water) enters into the absorber at 312 K and 0.112 MPa and at a flow rate of 93293.6 kg/h. The CO₂ concentration at the output of the absorber reaches less than 1% v/v. The amine-based water solution (low-concentration solvent) is fed from the upper part of the column and is poured down in the opposite direction of the flow, separating the CO₂ from other combustion gases. The dense CO₂ solution flows out of the column from the lower part and is pumped into the thermal converter to be directed to the purification tower.

The input and output temperatures for the dense CO₂ solution in the thermal converter are 328 k and 361 k, respectively. The MEA absorption tower is theoretically simulated in 35 trays. In the purification section, a solution with a high CO₂ concentration is produced in the rebuilder by reheating. The solution with low CO₂ concentration flows out of the lower

part of the purification tower and is sent to the thermal converter and enters the absorption tower after cooling.

The input and output temperatures for the solution with low CO₂ concentration in the heat transfer process are 389 k and 355 k, respectively. A MEA purification tower with 12 trays is theoretically simulated. The vapors of the condensed tower are used for the reproduction of CO₂. The CO₂ is compressed with a multi-stage compressor, cooled, and condensed, and is preserved with a cooling system with ammonium nitrate. The CO₂ is re-purified with a washer to remove any SO₂ impurity and is passed through several drying and absorption beds to meet the required standards for storing and transportation. The cooling water for the process is anticipated in advance. The amount of solvent with low concentration is highly effective for energy calculations of the process, which is optimized by altering the volumetric flow rate of the MEA. The Operating conditions of input and output flows are shown in Table 3.

Electrolysis of water

This part simulates the creation of the hydrogen required for downward methanol production. Water was supplied into an equilibrium reactor, which was utilized to replicate the alkaline electrolyzer's constant operation at its nominal point of 10.26 MW, the most mature, available, and inexpensive low-temperature water electrolysis technology at commercial scale [52,59]. Pumped and heated tap water at $P = 1$ atm and $T = 15$ °C (stream 101 in Fig. 3) to feed the electrolyzer at 70 °C and 10 bar. Hydrogen is formed by water electrolysis under three-phase conditions:

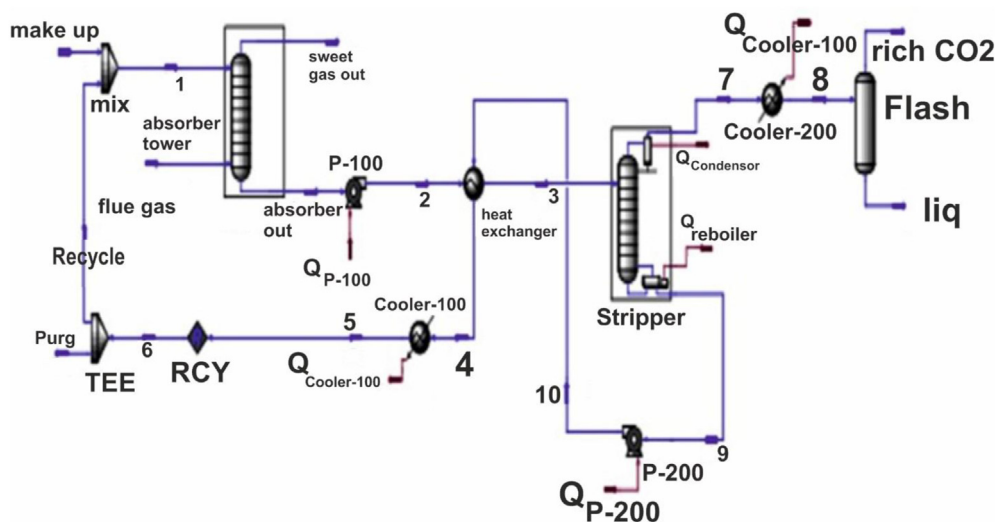
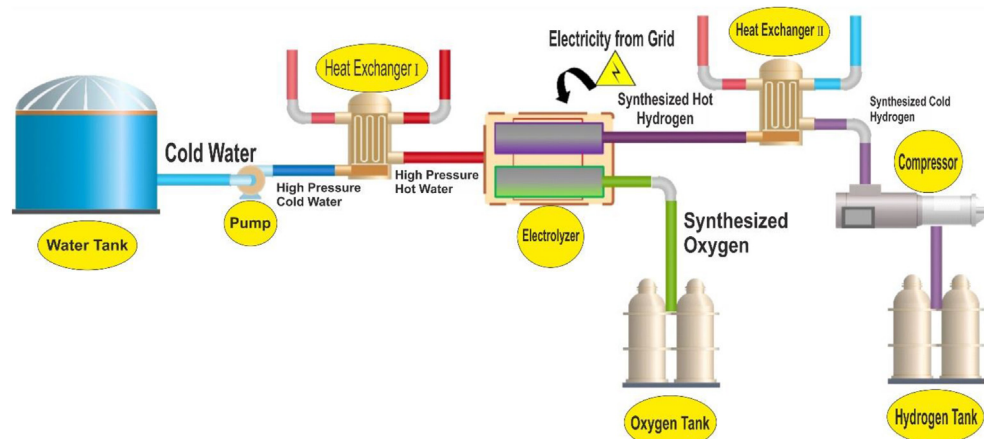


Fig. 2 – Simulation of CO₂ absorption process by MEA solvent. The process involves a column for separating CO₂ from combustion gas using MEA with a mass concentration of 29%. The molar fractions of the input and output flows are shown in Table 2.

Table 3 – Operating conditions of input and output flows.

Name	Vapor fraction	Temperature (°C)	Pressure (KPa)	Mass flow rate (kg/h)
Flue gas in	0.9737	39.85	112	93270.96
Sweet gas out	1	35.01	101	72246.28
Absorbent out	0	53.69	112	490476.6
1	0	35.02	136	463674.6
2	0	53.74	239	490476.6
3	0	87.85	239	490476.6
Stripper up	1	60.60	143	17153.28
Stripper bottom	0	114.4	164	473,502
4	0	81.05	136	473,502
5	0	34.85	136	473,502
6	0	34.85	136	473,502
8	0.8815	30	143	17153.28
10	0	114.4	175	473,502
Liq	0	30	143	910.9702
Rich CO ₂	1	30	143	16242.012
Recycle	0	34.85	136	426151.8
Purg	0	34.85	136	47350.2
Make up	0	34.85	136	43478.8

**Fig. 3 – The schematic of electrolysis of water.**

2778.09 kg/h of hydrogen (supposing a specific energy consumption for electrolysis of 5.4 kW h/Nm³ H₂ [41]) and 17,704.33 kg/h of oxygen, divided through a separator that simulates the electrolyte interposed between the two electrodes of a real alkaline cell in the model. Extra water was recirculated back to the reactor from the exit (i.e., to the stack inlet). As shown in Fig. 3, high-purity hydrogen was created, while pure oxygen was vented; nevertheless, it can be sold to boost the plant's income. Finally, at the input of the methanol synthesis unit, the produced hydrogen was cooled and compressed to the working conditions required.

Simulation of renewable methanol production with CO₂ and hydrogen synthesis

For the proposed renewable methanol production project, a total of 17,704.33 kg/h of CO₂ (at a temperature of 25 °C and a pressure of 1 bar) and 2778.09 kg/h of hydrogen (at a temperature of 25 °C and a pressure of 30 bar) is considered (Fig. 4).

The CO₂ is compressed to 78 bar by four serial compressors and the hydrogen is compressed to 78 bar in one stage. The two feedstock are mixed (MIX₁) and then mixed again with the recycled flow (MIX₂), which is then heated to 210 °C in a conversion (HX₄) and is fed into an adiabatic fixed-bed reactor. The conversion rate in this reactor is considered 33%. The output gases of the reactor (DIV₁) are divided into two streams. The first stream (60% of the primary stream) is used for heating the fresh feedstock (HX₄) and the second stream is used in the reboiler (DT1REB) for heating the distillation column (HX₅). The above streams are mixed again (MIX₃) and are cooled to 35 °C in the HX₆ converter with water. The water and methanol that have condensed in the HX₆ converter are separated from the unreacted gases in a drum (KO₁). A fraction of the unreacted gases (1%) is removed from the process to prevent the collection of inert gases and byproducts. The output liquid stream from the knock-out drum (KO₁), which is called raw methanol, contains methanol, water, and residual solution gases. The raw methanol pressure in two valves (VLV₁ and VLV₂) reaches 1.2 bar. The

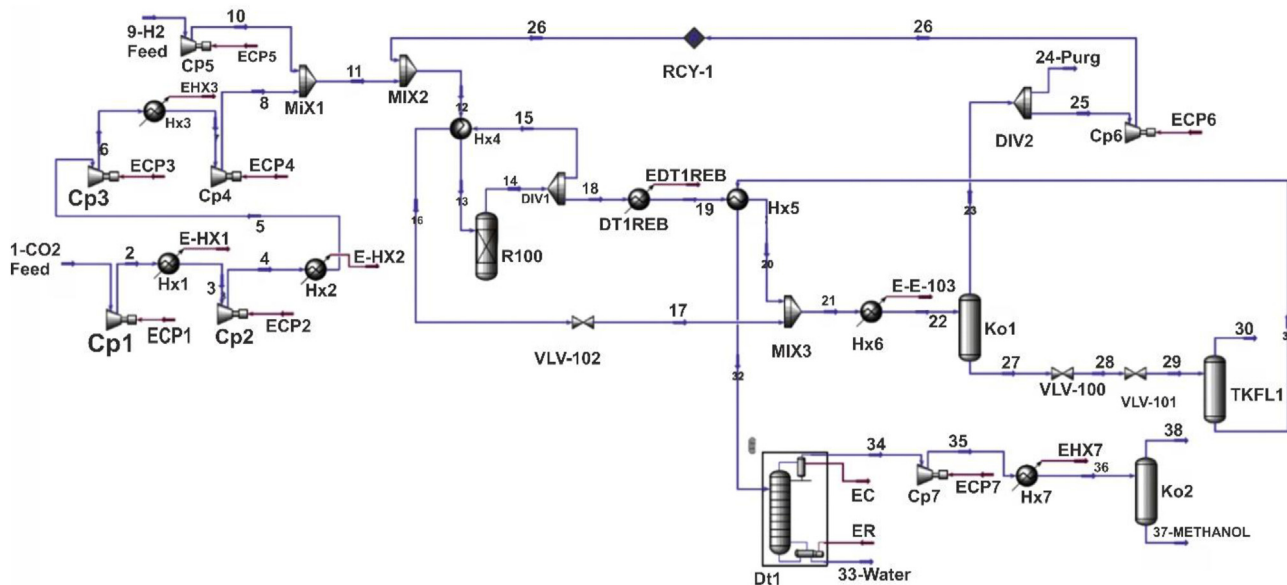


Fig. 4 – Simulation of methanol production process from CO₂ and H₂.

residual gases are then completely separated in the flash tank (TKF₁). The remaining stream in the HX₅ converter is heated to 80 °C and then is sent to the distillation column. Water containing a tiny amount of methanol is out from the bottom of the column with a temperature of 102. °C. Methanol in the form of gas containing a tiny amount of water and some unreacted gases is out from the top of the column at a pressure of 1 bar and a temperature of 64 °C. The methanol is then compressed in the CP₇ compressor and is cooled in the HX₈ converter to 40 °C. The overall mass balance of the outputs of the process of methanol production from CO₂ and hydrogen is presented in Table 4.

Results and discussion

Thermal and electrical energy consumption of the CO₂ separation and methanol production units

The electricity and thermal energy consumption of the CO₂ separation and methanol production units are detailed in Table 5.

A comparison of the energy consumption, efficiency, and CO₂ emissions of the traditional and renewable methanol production methods

According to Table 6, the consumption of electrical energy in the renewable methanol production method is approximately 1.045 times as much as in the traditional method of methanol production, which is in large part due to the electricity consumption of the CO₂ separation unit of the renewable methanol plant. However, the consumption of thermal energy in the traditional methanol production method is almost 2.5 times more than that of the renewable method, which is due to the use of natural gas as the feedstock for conventional methanol production. In addition, as for the environmental aspect of the two methods, as expected, the amount of direct and indirect CO₂ emissions from the renewable methanol production process is almost one-third of those from the conventional method. Furthermore, the environmental efficiency of renewable methanol production is 2 times more than that of the conventional method, while the energy efficiency of the conventional method is 1.5 times more than the renewable method.

Table 4 – The overall mass balance of the outputs of the process of methanol production from CO₂ and H₂.

	Purged	30	38	Water	Methanol	Total
Mass fraction						
CO ₂	0.187102	0.769398	0.639021	1.00E-30	4.74E-04	
H ₂ O	4.57E-04	1.34E-02	7.82E-07	1	6.69E-06	
Methanol	2.11E-03	6.73E-02	0.355013	2.15E-11	0.999519	
Oxygen	0	0	0	0	0	
Hydrogen	0.810334	0.149925	5.96E-03	9.99E-31	6.02E-08	
Mass Flow rate (kg/hr)	409.64	307.23	19700.61	7105.35	12625.37	20482.43

Table 5 – The energy consumption of the CO₂ separation and methanol production units.

CO ₂ separation unit	Energy Balance
Regeneration column	16,394,945 m ³ ^a
Absorption column	12,395,892 m ³ ^a
Total thermal energy consumption (N.G)	28,790,837 m ³
Auxiliary	251,125 MW h
CO ₂ compression	6350 MW h ^a
Total electrical energy consumption	257,475 MW h
Methanol unit	
Thermal energy outlet based on the heat of reaction (N.G)	4,616,480 m ³
Auxiliary	28,613 MW h
H ₂ +CO ₂ compression (from 30 up to 80 bar)	7890 MW h ^b
Total electrical energy consumption	36,503 MW h
Total thermal energy consumption (CO ₂ separation and Methanol units)	33,407,317 m ³
Total electrical energy consumption (CO ₂ separation and Methanol units)	293,978 MW h

^a The specific heat demands for amine regeneration are 4.17 MJ_{th}/kgCO₂. Considering a CO₂ capture rate of 85%, 44 kWh_{el} per ton of CO₂ are consumed for feed compression and 3.2 GJ_{th} per ton of CO₂ are employed for solvent regeneration [39].

^b The specific power consumption for H₂ compression is 78.9 kWh_{el}/ton MeOH [39].

Table 6 – A comparison of the energy consumption, efficiency, and CO₂ emissions of the traditional and renewable methanol production methods.

	Conventional methanol	Renewable methanol
Electrical energy (MWh/t _{MeOH})	2.811	2.939
Thermal energy (m ³ /t _{MeOH})	866 ¹	334.07
Direct CO ₂ emissions (tCO ₂ /tMeOH)	1.050	0.092
Indirect CO ₂ emissions (tCO ₂ /tMeOH)	1.813	0.733
Total CO ₂ emissions (tCO ₂ /tMeOH)	2.863	0.825
Efficiency		
μ	77.5	53.68
Cc	34.92	69.28

Economic analysis results

Investment cost estimation

Investment cost estimation for construction of renewable methanol production plant. According to Table 7, the biggest of the equipment purchase cost belongs to the methanol production process. The reactor, compressors, and distillation column have the biggest share of the cost, respectively, while the smallest share of the equipment cost belongs to the pumps.

The investment cost for the construction of a renewable methanol production plant is presented in Table 8. As an interdisciplinary study with experts from energy enterprises and higher institutions, the present research team obtained data from China and the Middle East. However, considering

Table 7 – Cost breakdown of purchased equipment renewable methanol production processes.

Equipment	Cost (\$)
Electrolyzer unit	16,765,884
CO ₂ separation unit	
Absorber- tower	1,031,324
Cooler 1	219,775
Cooler 2	778,854
Flash	46,589
Heat exchanger	139,767
Pump 1	31,167
Pump 2	29,237
Stripper-tower	801,026
PEC	3,135,666
Methanol unit	
Cooler 1	70,486
Cooler 2	249,793
Heat exchangers	380,874
Methanol distillations	3,903,918
Methanol reactor	17,310,772
Compressors	4,058,552
Separators	139,767
PEC	26,114,162
SUM	46,015,712

the data integrity, the data from Iraq is used as a case here. According to this table, the total cost of the construction of a renewable methanol production plant is estimated at \$77.67 million, with the equipment purchase cost accounting for the largest share.

Investment cost estimation for construction of conventional methanol production plant. According to data from the Office of Economic Studies of the Islamic Parliament Research

Table 8 – The investment of renewable methanol production plant.

Item	Cost (\$)
Direct Costs	
Purchased Equipment	46,015,712
Installation (insulation + painting),10%	4,601,571
Purchased Equipment	
Instrumentation & controls, installed	181,020
Piping, installed	663,740
Electrical, installed	110,623
Building, process & auxiliary	181,020
Service facilities & yard improvement, 3%	1,380,471
Purchased Equipment	
Land	60,340
Sum Direct Costs	53,194,497
Indirect Costs	
Engineering & supervision	331,870
Construction expense & contractors feed	211,190
Contengency fee, 30% Purchased	13,804,714
Equipment	
Sum Indirect Costs	14,347,774
Fixed-Capital Investment	67,542,271
Working Capital, 15% Fixed-Capital	10,131,341
Investment	
Total Capital Investment	77,673,612

Table 9 – Estimation of operating costs of CO₂ to methanol conversion unit.

Item	million dollars
Electrical energy consumption	0.822 ^a
Thermal energy consumption (natural gas)	3.341 ^b
Cost of catalyst	0.071
Cost of water	0.437
Direct wages	1.280
Depreciation	8.848
O & M cost	1.485
Cost of sales and marketing (1% Total Gross Sales)	0.25
Total production cost	16.534

^a The average subsidized electricity price in the industry, mines, and petrochemical sector is \$2.796 per MWh.

^b The price of natural gas feed for methanol production is about 10 cents per cubic meter.

Center, the total cost of the construction of a conventional methanol production plant is \$46.807 million.

Operational cost estimation

Cost of CO₂-to-methanol conversion unit. The breakdown of the operational cost of renewable methanol production is presented in Table 9, according to which the total operational cost of renewable methanol production totals \$16.534 million.

According to Fig. 5, depreciation accounts for the biggest [42] share of the operational cost of renewable methanol production. The cost of natural gas as the feedstock and the maintenance cost come second and third, respectively. The smallest share belongs to the cost of catalysts.

Cost of natural gas-to-methanol conversion. The breakdown of the operational cost of methanol production with the

Table 10 – Estimation of operating costs of natural gas to methanol conversion unit.

Item	million dollars
Cost of natural gas	6.168 ^a
Cost of catalyst	0.710
Cost of water	0.200
Direct wages	1.280
Electrical	0.786 ^b
Depreciation	4.424
O & M cost	0.594
Cost of sales and marketing (1% Total Gross Sales)	0.250
Total production cost	14.412

^a The average subsidized electricity price in the industry, mines, and petrochemical sector is \$2.796 per MWh.

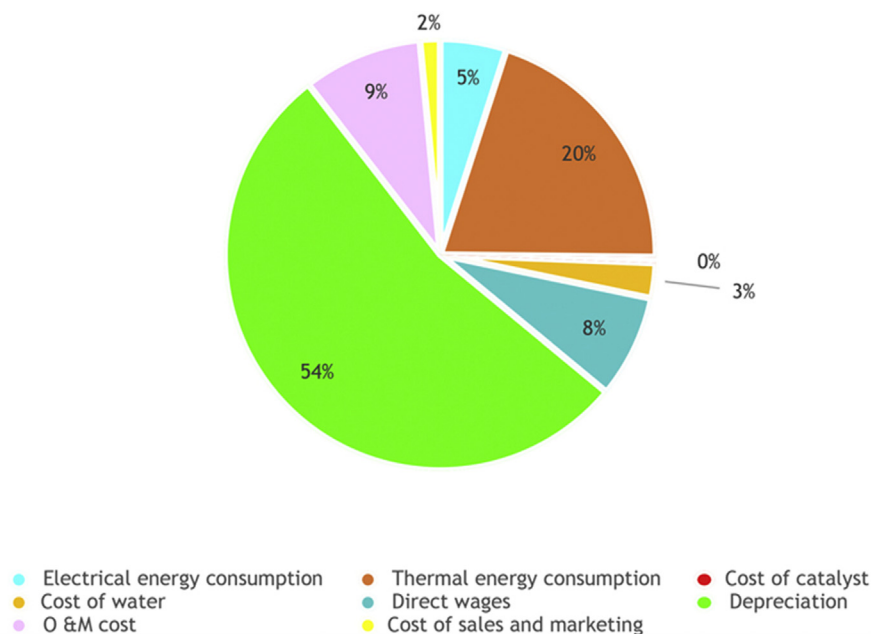
^b The price of natural gas feed for methanol production is about 10 cents per cubic meter.

conventional method is shown in Table 10, according to which the total operational cost of conventional methanol production amounts to \$14.412 million.

According to Fig. 6, the feedstock (natural gas) and depreciation carry the highest operational costs among other items. The lowest share of the operational cost belongs to water consumption.

A comparison of the economic performance of the traditional and renewable methanol production methods

According to Table 11, the investment cost and production cost of renewable methanol production are roughly 1.07 and 1.147 times as much as that of the traditional method, respectively. All in all, given that the IRR of the renewable methanol production is 17.27%, which is lower than the discount rate of 18% approved by the Central Bank, this method is less economically viable than the conventional production method.

**Fig. 5 – Operating costs of renewable methanol production.**

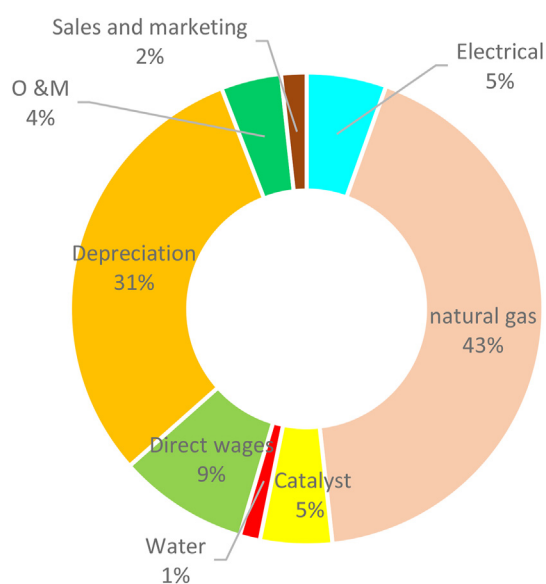


Fig. 6 – Operating costs of natural gas to methanol.

Sensitivity analysis

The effect of methanol price volatilities on the IRR. Three price scenarios including the pessimistic scenario of \$200 per ton in 2020, realistic scenario of \$250 per ton (average price for 2020 and January 2021), and optimistic scenario of \$310 per ton in January 2021) are considered and the effect of each of them on the IRR and the payback period in both conventional and renewable methanol production are evaluated.

According to Fig. 7, if methanol prices drop to \$200 per ton, both conventional and renewable methanol production will lose their economic viability, but if it rises to \$310 per ton, renewable methanol production will be highly economically viable.

The impact of real (unsubsidized) electricity prices on the IRR. The average subsidized electricity price in the industry, mines, and petrochemical sector is \$2.796 per MWh while the real and unregulated price is 6056.7 \$25.238 per MWh. In other words, the government of supplies the electricity needed for the industry, mines, and petrochemical sector with a massive subsidy of % 902.646 (. Accordingly, three price scenarios are presented for the electricity consumption

Table 11 – Comparison of the economic performance of the traditional and renewable methanol production methods.

Item	traditional methanol (million dollars)	renewable methanol (million dollars)
Fixed investment cost	46.807	77.674
Total production cost	14.412	16.534
Methanol	25	25
O ₂	0	6.3
Total Gross Sales	25	31.3
Total net profit	10.588	14.746
Tax (9% Total net profit)	0.953	1.327
Total net profit after tax	9.635	13.418
Economic Index		
IRR	20.58%	17.27%
Payback period	4.858 years	5.788 years

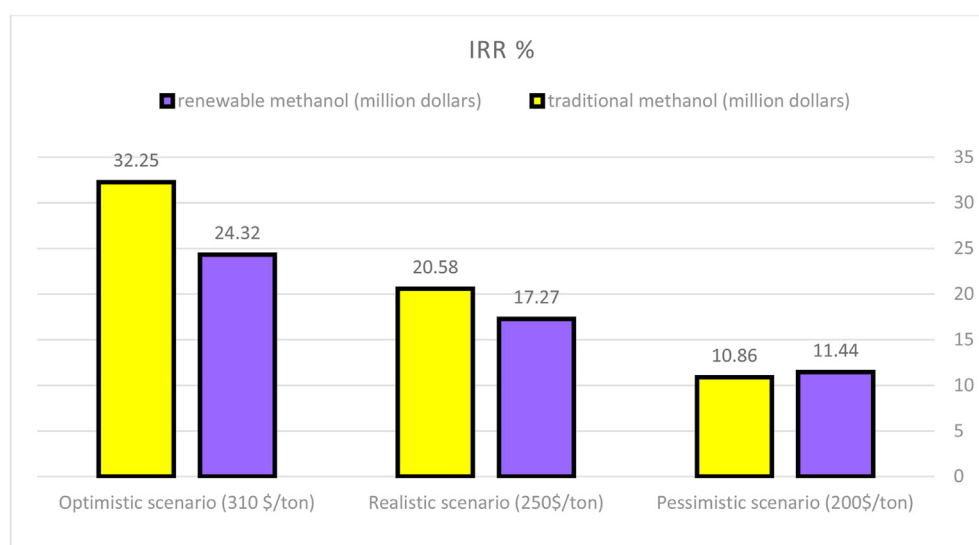


Fig. 7 – The effect of methanol price volatilities on the IRR.

of the methanol plants, which include the pessimistic scenario (based on the price of electricity export to, i.e. \$120 per MWh), realistic scenario (unsubsidized price, i.e. \$25.238 per MWh), and optimistic scenario (current subsidized price, i.e. \$2.796 per MWh).

As shown in Fig. 8, if the electricity needed for the methanol production plants is provided at the same rates as the export rates or the unsubsidized rate, both conventional and renewable methanol production will lose their economic viability. This means that the methanol production industry is highly dependent on the support of the government.

The impact of real (unsubsidized) natural gas prices on the IRR. Three price scenarios are presented for the natural gas consumption of the methanol production plants, which include

the pessimistic scenario (the price of export gas to Turkey, i.e. \$0.33 per m³), realistic scenario (current unsubsidized price, i.e. \$0.1 per m³), and optimistic scenario (subsidized price, i.e. \$0.07 per m³).

As shown in Fig. 9, if natural gas is provided to the conventional and renewable methanol production plants at the same price of export to Turkey, both conventional and renewable methanol plants will be economically unviable (the IRR of the conventional plants will plunge). But if the government supplies the plants with subsidized natural gas, even the renewable methanol production will have economic viability.

The impact of carbon emission taxation on the IRR. At the time of writing this paper, no tax measures have yet been introduced against carbon emissions. With that in mind, three

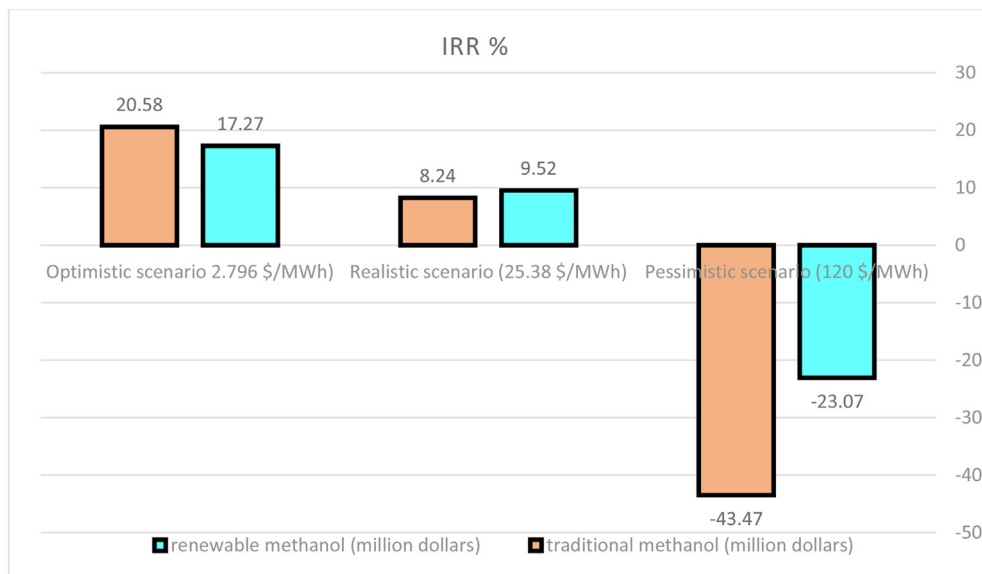


Fig. 8 – The impact of real (unsubsidized) electricity prices on the IRR.

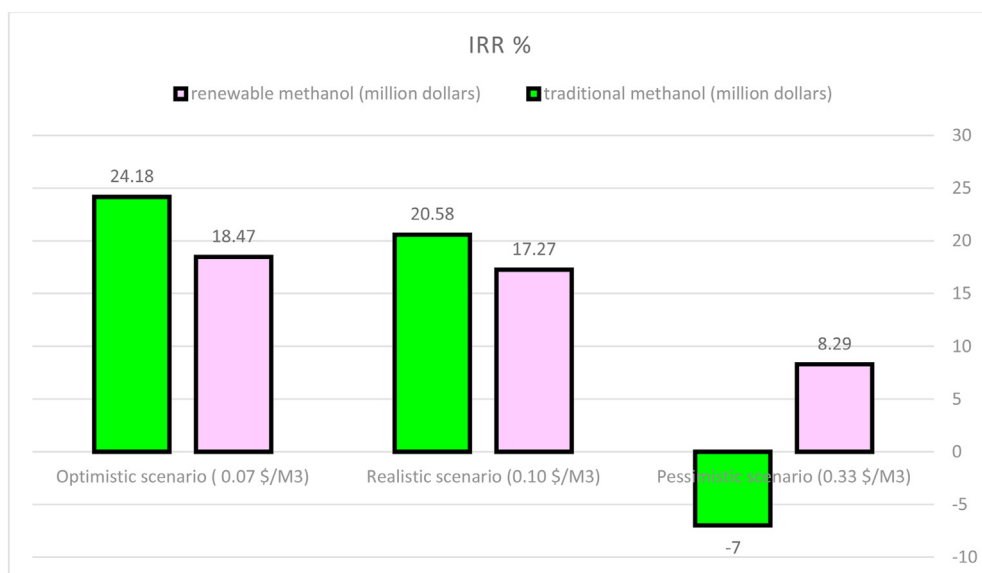


Fig. 9 – The impact of real (unsubsidized) natural gas prices on the IRR.

scenarios for tax on carbon emissions produced by methanol production plants are presented here, which are as follows: pessimistic scenario (\$35 per ton), realistic scenario (\$14 per ton), and optimistic scenario (current situation, i.e. no tax). As shown in Fig. 10, both traditional and renewable methanol production plants will be economically unviable if taxes are imposed on carbon emissions.

The impact of oxygen price on the IRR. As a by-product of water electrolysis, a large amount of oxygen is created (8 kg per kg of hydrogen). When compressed or liquefied and stored in a tube

trailer (with purchasing and operating costs that are not taken into account in our techno-economic analysis), oxygen can be used for a variety of industrial and manufacturing applications, resulting in a second source of income in the cash flow analysis.

Three price scenarios are presented for the oxygen production from the water electrolysis, which include the pessimistic scenario (\$15.45 per m³), realistic scenario (current price, i.e. \$ 40.83 per m³), and optimistic scenario (\$51.03 per m³). As shown in Fig. 11, according to the pessimistic scenario and realistic scenario, renewable methanol plants will be economically unviable.

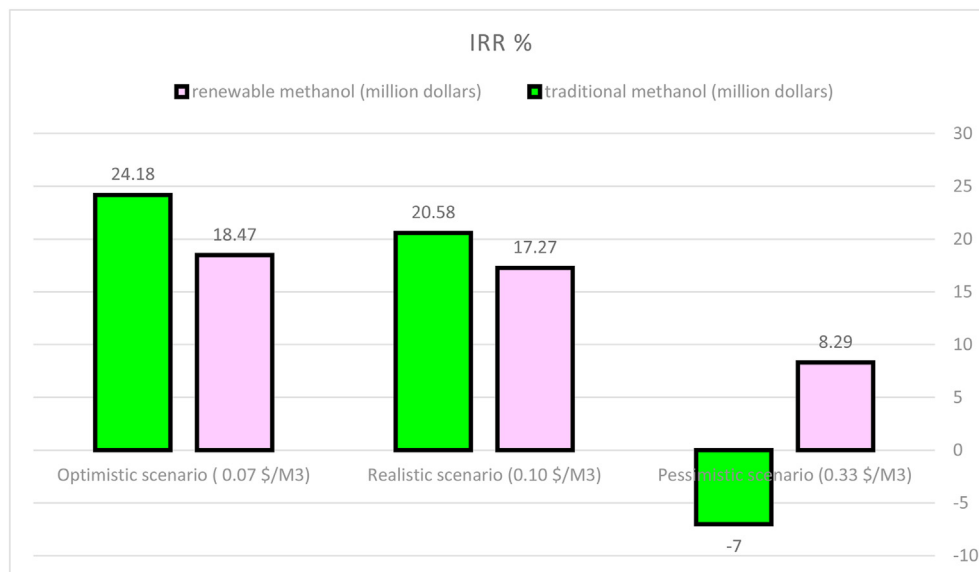


Fig. 10 – The impact of carbon emission taxation on the IRR.

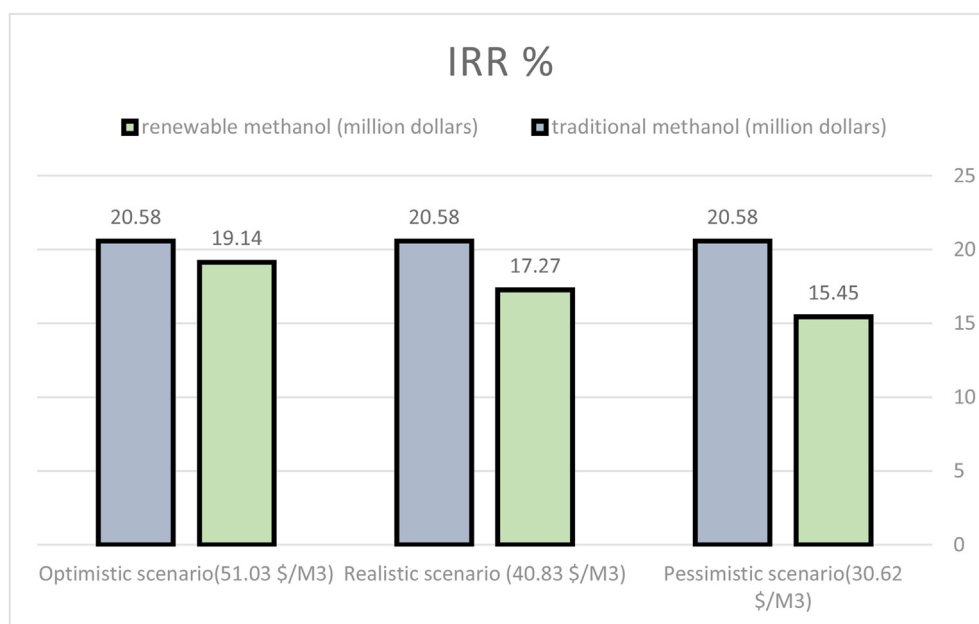


Fig. 11 – The impact of oxygen price on the IRR.

Conclusion

The purpose of this study was to compare and contrast the conventional and renewable methanol production methods through a thermoeconomic analysis. Synthesis gas is produced from a compound of hydrogen from water electrolysis and CO₂ from a gas power plant and then it is converted into renewable methanol by going through a few stages. The methanol production plant with a capacity of 100,000 tons per year from renewable sources was simulated using Lurgi technology and ASPEN HYSYS V.10 and then was economically analyzed using COMFAR III.

- Electricity consumption

The electricity consumption of renewable methanol production is roughly 1.045 times as much as that of the conventional method.

- Thermal energy consumption

The thermal energy consumption of the conventional methanol production method is roughly 2.5 times as much as that of the renewable method.

- Direct and indirect CO₂ emissions

The total direct and indirect CO₂ emissions from the renewable methanol production process are about one-third of those from conventional methanol production.

- Investment cost

The total cost of setting up a renewable production plant with an annual capacity of 100,000 tons is \$50.1 million. This is while the total capital cost for the construction of a conventional methanol production plant with a capacity of 100,000 tons per year is \$46.807 million.

- Operational costs

The total operational cost of renewable methanol production is \$16.534 million. This is while the operational cost in the conventional production method is \$14.412 million.

- Impact of real electricity price (unsubsidized) on IRR

If the required electricity for conventional and renewable methanol production plants is supplied without state subsidies or at the price of its export, both conventional and renewable methanol production plants will lose their economic viability.

- Impact of real natural gas price (unsubsidized) on IRR

If the required natural gas for renewable and conventional methanol production plants is supplied at the price of its export to Turkey, both conventional and renewable methanol production plants will lose their economic viability (the IRR of

conventional methanol production plants will plummet). But if the supplied natural gas for conventional and renewable methanol plants is charged at the subsidized price, then even renewable methanol production will be economically viable.

- Impact of carbon tax on IRR

The potential imposition of tax on carbon emissions will make both conventional and renewable methanol production plants economically unviable (the IRR of conventional methanol production plants will plummet).

- Impact of oxygen price on IRR

According to the pessimistic scenario and realistic scenario, renewable methanol plants will be economically unviable. All in all, given that the Internal Rate of Return of a renewable methanol production plant is 17.27% which is lower than the discount rate of 18% approved by the Central Bank, this method is less economically viable than the conventional production method. However, if the government slightly raises the subsidized electricity and gas prices in the petrochemical industry, a renewable methanol production plant will be highly economically attractive. To realize this, countries can establish special high-tech parks to give favourable subsidy policies and carbon tax exemption just like advocated by China [53–55]. Previous studies has proved high-tech parks can test the effectiveness of tax exemption and subsidy policies in a limited scale and provide a reference for larger scale implementation [56]. Especially in the green methanol production, the cooperation between the energy producers like Middle East countries and energy consumers like China to establish high-tech parks to realize technology incubation is conducive to jointly coping with the pressure brought by energy transition [57].

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Nomenclature

RES	Renewable Energy Sources
PV	Photo Voltaic
RWGS	Reverse Water Gas Shift
MEA	Mono Ethanol Amine
IRR	Internal Rate-of-Return

NPV	Net Present Value
N.G	Natural Gas
PEC	Purchase Equipment Cost
O & M	Operations and maintenance Cost
CEPCI	Chemical Engineering Plant Cost Index

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